Linux for DevOps

Lesson 6: Networking & Security Essentials

Networking & Security Essentials - Study Notes

Key Concepts

1. **IP Configuration** – Assigning static or dynamic IP addresses, subnet masks, gateways, and DNS servers.

2. **SSH (Secure Shell)** – Remote login protocol that encrypts traffic, uses key based authentication, and supports port forwarding.

3. **firewalld & firewall cmd** – A dynamic firewall manager for Linux that uses zones, services, and rich rules.

4. **Basic Firewall Rules** – Allowing/denying traffic by protocol, port, source/destination IP, and state.

5. **Security Best Practices** – Least privilege, strong key management, regular updates, and logging.

Summary

Networking & Security Essentials covers the foundational steps to secure a Linux host: configuring network interfaces, enabling secure remote access with SSH, and hardening the system with `firewalld`. First, you learn how to set static IPs or use DHCP, ensuring the machine can communicate on the network. Next, SSH is configured for secure remote administration, emphasizing key based authentication over password logins. Finally, `firewalld` is introduced as a flexible, zone based firewall that allows you to define rules for inbound/outbound traffic, block unwanted services, and log suspicious activity.

By the end of this lesson you will be able to:

• Assign a static IP to a network interface and verify connectivity.

• Generate SSH key pairs, configure `sshd_config`, and restrict access.

• Use `firewall-cmd` to enable/disable services, open/close ports, and apply rich rules.

• Understand how to test firewall rules with `telnet` or `nc` and review logs in `/var/log/messages` or `journalctl`.

Important Points

• **IP Configuration**

- `/etc/sysconfig/network-scripts/ifcfg-eth0` (RedHat/CentOS) or `/etc/netplan/*.yaml` (Ubuntu) for static IPs.
- `ip addr add`, `ip route add`, and `systemctl restart network` for quick changes.

• **SSH**

- Generate keys: `ssh-keygen -t ed25519`.
- Disable password auth: `PasswordAuthentication no` in `/etc/ssh/sshd_config`.
- Restart SSH: `systemctl restart sshd`.

- Use `ssh -i ~/.ssh/id_ed25519 user@host`.
  - **firewalld**
- Zones: `public`, `work`, `drop`, etc.
- Add service: `firewall-cmd --zone=public --add-service=http --permanent`.
- Open port: `firewall-cmd --add-port=8080/tcp --permanent`.
- Reload rules: `firewall-cmd --reload`.
- Rich rules for complex logic:
  ```bash
  firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="192.168.1.0/24" port port="22" protocol="tcp" accept'
  ```
  - **Logging & Monitoring**
- Check firewall logs: `journalctl -u firewalld`.
- SSH logs: `/var/log/secure` or `journalctl -u sshd`.
  - **Security Hygiene**
- Keep packages up to date: `yum update` / `apt update && apt upgrade`.
- Use `fail2ban` to block repeated failed SSH attempts.
- Regularly review firewall rules with `firewall-cmd --list-all`.

Examples

1. Setting a Static IP on CentOS 8

```bash
# /etc/sysconfig/network-scripts/ifcfg-eth0
DEVICE=eth0
BOOTPROTO=static
ONBOOT=yes
IPADDR=192.168.10.10
NETMASK=255.255.255.0
GATEWAY=192.168.10.1
DNS1=8.8.8.8
DNS2=8.8.4.4
```

Restart networking:

```bash
systemctl restart network
```

Verify:

```bash
ip addr show eth0
```

ping -c 4 192.168.10.1
```

**2. Restricting SSH to Key Only Access**

```bash

# Generate key on client

ssh-keygen -t ed25519 -C "admin@example.com"

# Copy public key to server

ssh-copy-id -i ~/.ssh/id_ed25519.pub user@192.168.10.10

# On server, edit /etc/ssh/sshd_config

PasswordAuthentication no
PermitRootLogin no
ChallengeResponseAuthentication no

# Restart SSH

systemctl restart sshd
```

Now only key based logins are accepted; password attempts will fail.

Quick Review

1. What command would you use to assign a static IP to `eth0` on a Debian system using Netplan?
2. How do you disable password authentication for SSH while keeping key based login active?
3. Explain the difference between a zone and a service in `firewalld`.
4. Write a `firewall-cmd` command to permanently allow HTTPS traffic in the `public` zone.
5. Which log file would you check to see failed SSH login attempts?

Further Reading

- **Advanced SSH Features** – X11 forwarding, agent forwarding, and port forwarding.
- **firewalld Rich Rules** – Conditional rules, logging, and reject actions.
- **SELinux/AppArmor** – Mandatory Access Control for Linux.
- **Network Address Translation (NAT)** – Using `iptables` or `firewalld` for port forwarding.
- **Intrusion Detection Systems** – `fail2ban`, `Snort`, or `Suricata`.
- **IPv6 Basics** – Configuring IPv6 addresses and firewall rules.
```